

Dhruv Deshpande

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Education

B.Tech in Computer Science (Specialization: Data Science)

Sardar Patel Institute of Technology, Mumbai

2022 – 2026

CGPA: 8.33

CBSE Senior Secondary (12th), Ryan International School

2022

Percentage: 95.20%

CBSE Secondary (10th), Ryan International School

2020

Percentage: 98%

Profile

Passionate and dedicated student with a strong interest in Data Structures and Algorithms (DSA) and the mathematical foundations of Machine Learning. Exploring applications of analytics and modeling in finance. Excellent team player with strong communication and leadership skills.

Technical Skills

Languages: Python, SQL, Java, C

Libraries/Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, React.js, Node.js, Express.js

Tools & Platforms: Power BI, Excel, MongoDB, Jupyter Notebook, Git

Other: Strong mathematical foundation (Probability and Statistics)

Projects

FinTrack – Inflation Forecasting and Financial Insight System

2025

Built a hybrid inflation forecasting system using LSTM for temporal modeling and XGBoost for accurate regression, with K-Means-based user clustering for segmented financial analysis. Used FinLDA on data and sentiment to generate interpretable market insights, integrating time-series, clustering, and topic modeling for explainable and personalized financial predictions.

Risk Scope – Credit Risk Classification Model

2025

Developed a scalable credit default classification pipeline using Python and Scikit-learn, implementing logistic regression with L1 and L2 regularization for financial modeling. Constructed key financial and macroeconomic features (e.g., Debt-to-Equity, Current Ratio, ROA) and applied correlation-based filtering alongside L1-penalty feature selection to optimize model inputs. Assessed model performance using ROC curves and precision-recall analysis; delivered insights through interactive Power BI dashboards.

Neuro-Think – Reinforcement Learning-Based Chess Engine

2025

Built a chess engine from scratch using reinforcement learning, training via self-play with Monte Carlo Tree Search (MCTS). Integrated a deep neural network for state evaluation and policy prediction, improving move accuracy and learning efficiency.

Achievements

Runner-up – CodeHousie, a logic-driven coding challenge by CSI-SPIT.

Cleared JEE Advanced 2022.

Champion of the Inter-College Carrom Tournament.

Certifications

Supervised ML, Coursera – Score: 100%

Apr 2025

Neural Networks, Coursera – Score: 96%

Apr 2025

Unsupervised Learning + RL, Coursera – Score: 100%

May 2025

Extra-Curricular Activities

Chess.com Rapid Rating: 1700; winner in multiple inter-college events.

Champion of the Inter-College Carrom Tournament.